

“EV BATTERIES Are GETTING BIGGER And So Are Charging Power Requirements”

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Basic understanding of modern power electronics devices is essential in automotive or the electronics design/manufacturing industry due to their revolutionary role in energy conservation for high-efficiency automotive and non-automotive applications. Niranjan S. Nayak, business head, Delta Electronics India Pvt Ltd, discusses the technical requirements for developing various such solutions, with a focus on electric vehicle (EV) automotive applications, with Ayushee Sharma

Q. How do DC/DC converters used in automotive differ from those in non-automotive transportation?

A. Operational requirements, environmental conditions and safety requirements differ for automotive and non-automotive vehicles. There are well-defined standards to which automotive components have to conform, whereas non-automotive vehicles have to conform to standards governing their end application.

As far as safety is concerned, DC/DC converters used in automotive are designed as per ISO 26262 (automotive-specific functional safety standard), whereas non-automotive vehicles, used in industrial applications, should also conform with EN 62061 and ISO13849-1.

Q. What are the various tools used in designing DC/DC converters?

A. In the automotive sector, components are custom-designed based on requirements of OEMs. IBM Rational DOORS is a well-known tool for requirement management.

Thereafter, UML tools like Sparx Enterprise Architect or IBM Rational Rhapsody can be used for system architecture design. For electrical circuit design, tools like MATLAB and PSpice can be used. This is followed by schematic and PCB layout design, and tools like Altium and Cadstar are useful in this phase.

PTC Creo is widely-used in the automotive industry for mechanical design. AUTOSAR is a framework that pursues creating standardised software architecture for automotive embedded software. Tools from Mentor and Vector Informatik are used for this.

Q. What are the protection options for converters in harsh environments?

A. Protection is included for over-current, over-voltage and over-temperature. Apart from this, high-voltage (HV) plug-detection and crash-detection are examples of other protection features that are included specifically in automotive HV components.

For automotive domain, based on severity of hazard, ISO 26262 defines four ASIL (automotive safety integrity level), A, B, C and D, in increasing order of hazard. Protection systems defined above are designed to meet ASIL requirements.

To elaborate on complexity, over-current protection system designed to meet ASIL-D requirement has to have two current sensors and two independent channels to monitor current. Further, the MCU used should also meet ASIL-D requirement and so should the software.

Q. What are the considerations for designing EV/HEV powertrain solutions?

A. EV/HEV (hybrid EV) powertrain solutions are designed keeping consideration of performance, safety, size weight and cost. There are different types of EVs available in market, which differ in size, range, performance and cost. EV powertrains are designed to meet these conflicting requirements, and end-product is the result of optimisation with multiple conflicting objectives.

Q. What is the role of DC and AC filters in automotive applications?

A. AC and DC filters are one of the most critical and difficult-to-design parts in a car. In layman's terms, filters ensure that operation of one compo-

nent (for example, DC/DC) does not adversely affect other components (for example, infotainment) in a car. CISPR 25, ISO7637 and J1113/1 are some EMC standards that define limits of allowed electromagnetic interference (EMI) levels. Filters must be designed to meet these limits.

Q. What are the challenges involved in developing EV charging solutions?

A. EV batteries are getting bigger and so are charging power requirements. Initial EVs used 3.3kW onboard chargers (OBCs); today, high-end vehicles require 11kW or more power from OBCs. Moreover, due to size constraints, more OEMs are asking for integrated solutions called combo-chargers, which include OBC and DC/DC converter in one package.

A car has limited space to fit all components—with increasing power, size of charger is bound to increase. Therefore customers are asking for higher power density. This poses a challenge to companies like us to reduce size of bulky components by striving for highly-efficient and compactly-designed solutions. Today, our OBC have achieved power density level of 1.7kW/L and efficiency higher than 95 per cent. We are working towards achieving increasing power density of the level of 2kW/L.

Designing of chargers also includes significant software efforts as chargers have to interact with AC and DC supply equipment on one side and vehicle on the other. Charging protocols like CHAdeMO, GB/T and CCS must be implemented at software level within the OBC—based on OEM requirement, DC charging communication is handled by the OBC or another EV component. **EFY**